

TEAM INFINITY • PARALLAX HACKATHON

THE NEXUS

Multi-Objective Dynamic Workload Orchestration
for Edge-Core Cloud Architectures.

The Enterprise Edge **Crisis**



The Latency Trap

Static deployment models cannot adapt to unpredictable edge traffic surges. Human monitoring is fundamentally too slow, resulting in severe SLA latency violations and dropped requests.



The Cost Sinkhole

To prevent edge outages, enterprises centralize workloads or precautionary over-provision hardware. This results in massive redundant compute costs and idle server waste.



The Carbon Impact

Inefficient workload routing and over-provisioned data centers directly translate to significantly higher power consumption and an unacceptable carbon footprint.

Introducing Nexus Edge

PLATFORM OVERVIEW

Autonomous, AI-Driven Infrastructure.

An intelligent control plane that continuously evaluates network telemetry, node capacity, and energy usage in real time.

- ✓ **Distributed Telemetry:** Real-time metric aggregation across edge/core.
- ✓ **Multi-Objective Optimizer:** Balances latency, cost, and power constraints.
- ✓ **Zero-Downtime Migration:** Live workload shifting with zero state loss.

NEXUS INTELLIGENCE

90% CONFIDENCE

DERIVED

⊕ AUTONOMOUS ASSESSMENT

① WHAT HAPPENED

8 workloads across 7 online nodes, mean latency 20.0 ms. No SLA under pressure.

↗ WHAT WILL HAPPEN

No constraint is projected to breach inside the 90s horizon.

② WHY IT MATTERS

Nothing requires intervention. Tightest budget: AR-VR-01 with 5.4 ms of headroom.

💡 RECOMMENDED ACTION

Hold current placement. Continue observing.

⦿ EXPECTED RESULT

Steady state maintained.

Method: Statistical forecasting + deterministic weighted-sum optimisation. Telemetry is SIMULATED; energy and cost are ESTIMATED from published models.

System Architecture Core

Continuous Telemetry Pipeline

Low-overhead observability framework utilizing eBPF probes and Prometheus for real-time ingestion. Tracks end-to-end latency, available bandwidth, and node CPU/RAM/GPU utilization at the millisecond level.



SD-WAN Interconnect

High-throughput, low-latency SDN backbone linking edge sites for seamless stateful transfers.

Multi-Cluster Control

Unified control plane spanning heterogeneous hardware for standard lifecycle management.

Policy & SLA Enforcement Engine

Strict performance bounds formulation. If a microservice breaches the maximum 20ms response time threshold, the engine triggers proactive redistribution instantly.

Predictive Workload Forecasting

The analytics module anticipates regional traffic spikes and network congestion days in advance, shifting architecture from **reactive** scaling to **predictive** migration.



Autonomous Action

Zero-Touch Resolution

The orchestration engine requires no human approval to maintain stability. It acts as the air traffic controller for global data.

- ✓ Evaluates optimal nodes in real-time.
- ✓ Live-shifts containers with near-zero state transfer overhead.
- ✓ Maintains absolute SLA compliance across geographical zones.

CONTROL-PLANE TIMELINE		All	Forecast	SLA	Decisions	Routing	Migration	Topology	Failure	Scenario
12:11:00	ROUTING	Reroute completed: VIDEO-ANALYTICS-02								t47
VERIFIED — 19.0 ms → 19.2 ms (-1.1%), SLA AT_RISK → HEALTHY.										
12:10:58	SLA OK	SLA restored: VIDEO-ANALYTICS-02								t46
19.2 ms observed against a 45 ms limit (25.8 ms headroom). All constraints satisfied.										
12:10:57	ROUTING	Route installed: VIDEO-ANALYTICS-02								t44
Metro South → Edge Node 03 → Edge Node 05 → Edge Node 02										
12:10:55	ROUTING	Reroute requested: VIDEO-ANALYTICS-02								t42
Metro South → Edge Node 05 → Edge Node 02 ⇒ Metro South → Edge Node 03 → Edge Node 05 → Edge Node 02. Expected 19.2 ms end-to-end. Reroute VIDEO-ANALYTICS-02: 38.0 ms → 19.2 ms predicted, SLA restored.										
12:10:55	DECISION	Decision for VIDEO-ANALYTICS-02: REROUTE								t42
Reroute VIDEO-ANALYTICS-02: 38.0 ms → 19.2 ms predicted, SLA restored.										
12:10:54	ROUTING	Reroute completed: VIDEO-ANALYTICS-02								t41
NOT verified — 19.2 ms → 19.0 ms (+1.0%). The new path is installed but no measurable improvement was recorded.										
12:10:51	ROUTING	Route installed: VIDEO-ANALYTICS-02								t38
Metro South → Edge Node 05 → Edge Node 02										
12:10:51	FORECAST	Predicted capacity exhaustion: EDGE-02								t38
Memory 20% → 92% — probability 82% within 90s.										
12:10:49	ROUTING	Reroute requested: VIDEO-ANALYTICS-02								t37
Metro South → Edge Node 03 → Edge Node 05 → Edge Node 02 ⇒ Metro South → Edge Node 05 → Edge Node 02. Expected 19.0 ms end-to-end. Reroute VIDEO-ANALYTICS-02: 38.4 ms → 19.0 ms predicted, SLA restored.										
12:10:49	DECISION	Decision for VIDEO-ANALYTICS-02: REROUTE								t36
Reroute VIDEO-ANALYTICS-02: 38.4 ms → 19.0 ms predicted, SLA restored.										
12:10:48	ROUTING	Reroute completed: VIDEO-ANALYTICS-02								t35
NOT verified — 19.0 ms → 19.2 ms (-1.1%). The new path is installed but no measurable improvement was recorded.										
12:10:47	FORECAST	Predicted cpu overload: EDGE-02								t34
CPU 22% → 95% in 90s (linear_regression) — probability 76% within 90s.										

The Commercial Impact

Nexus transforms edge-core infrastructure from a reactive cost center into an optimized, self-healing ecosystem.

0ms

Unexpected Downtime

Hard SLA enforcement guarantees application performance metrics are met.

30 %

TCO Reduction

Eliminates the need for precautionary hardware over-provisioning at edge nodes.

⚡Min

Energy Waste

Power-aware placement algorithms actively minimize the overall carbon footprint.

Enterprise Security & Privacy



Live PII Scrubber

On-the-fly data masking meticulously protects sensitive customer information during all processes.



Zero-Trust mTLS

Strict architectural protocols ensuring completely secure, authenticated node-to-node communication.



Immutable Audit Trails

Creates perfect, tamper-proof records of every autonomous system decision for compliance teams.

The future is autonomous.

Thank you for your time. Open for
Questions.

Team Infinity

T Mohit Sai • Krishna Kumar Saravanan • V D Sanjayi